Sheikh Shihab Uddin

AI Engineering Internship Candidate | *Agentic AI Systems* | *LLM Tool Calling & MCP*
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PROFESSIONAL SUMMARY

Computer Science & Engineering student at BRAC University (expected 2028) focused on AI agents. Built a multi-provider LLM agent harness with MCP tool integrations, a network-isolated Docker code sandbox, Telegram human-approval gates, and a streaming web UI. Skilled in Python and TypeScript, with embedded robotics and technical video experience. Seeking an AI engineering internship to build reliable, secure agent systems.

EDUCATION

B.Sc. in Computer Science and Engineering | *BRAC University, Dhaka* *Expected 2028*
Relevant coursework: Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Advanced Calculus

PROJECTS

Multi-LLM Agent Harness with MCP & Sandboxed Execution | GitHub
Python, TypeScript, Next.js, Docker, Ollama, MCP, Telegram Bot API

- **Multi-LLM Orchestration:** Built a local-first agent with a streaming tool-calling loop and persistent session state, with automatic fallback across 5 LLM providers (Ollama, Gemini, Groq, OpenAI, Anthropic).
- **MCP Tool Integration:** Integrated 7 MCP servers (Time, Web Fetch, Memory, GitHub, Context7, Playwright, Sequential Thinking) with schema validation and isolated error handling, so one server failure never blocks the agent.
- **Secure Code Execution:** Sandboxed LLM-generated Python in Docker with no network, a read-only filesystem, 0.5 CPU / 256 MiB limits, and a 30-second timeout.
- **Human-in-the-Loop Safety:** Gated every data-mutating action behind a Telegram Approve/Deny prompt with a 60-second auto-timeout.
- **Web Interface:** Built a Next.js/TypeScript chat UI with live token streaming (SSE), a tool-activity log, and skill/MCP management; secured it with bearer-token auth and strict CORS, with mutating tools denied in web mode.
- **Evaluation:** Validated tool selection, context retrieval efficiency, and failure recovery with an offline suite of 84 tests (14 CLI-specific).

Encrypted Credential Vault & Password Manager | GitHub
Python, Cryptography

- Built a modular CLI password manager with symmetric encryption at rest, .env-based secret isolation, and documented secure API-key setup.

Autonomous Robotic Watercraft *Oct 2026*
Arduino, Embedded C/C++, Sensors

- Built an autonomous boat with remotely set waypoint navigation and sensor data collection using Arduino and motor drivers.
- Implemented motor-control logic and hardware-software communication for real-time sensor processing.

EXPERIENCE & LEADERSHIP

STEM & CS Tutor | *Dhaka* *Jan 2023 – Present*

- Taught 10+ SSC, HSC, and first-year university students in mathematics, physics, ICT, and computer science, turning topics like calculus and algorithm execution into structured lessons.

Active Member | *BRAC University Robotics Club (BURO)* *Oct 2026 – Present*

- Collaborated with hardware and software teams on embedded systems, sensor integration, and robotics prototypes.

TECHNICAL SKILLS

Agentic AI: LLM Agents, Tool Use / Function Calling, Model Context Protocol (MCP), ReAct Loops, Multi-LLM Orchestration, Human-in-the-Loop (HITL) Workflows, Agent Evaluation, Context Management
Languages & Web: Python (Asyncio, Pydantic, Aiohttp), TypeScript/JavaScript, React, Next.js, Tailwind CSS, Java, C/C++
Infrastructure & Security: Docker, Linux/Bash, Git/GitHub, uv, Sandboxed Code Execution, Secrets & Environment Management
Testing: unittest, Mocking, Offline Regression Testing

PROBLEM SOLVING

Codeforces: Solving algorithmic problems to strengthen data structures, algorithms, and contest-style problem solving. Codeforces

ADDITIONAL

Creative & Multimodal: 3D motion graphics, visual storytelling, and technical video editing; used to produce demo videos and architecture walkthroughs for agent projects. Creative portfolio
Languages: Bangla (Native), English (Professional Working Proficiency)